

Maximilian Saller, Ph.D.

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Date of Birth: 5/1/1991 • Citizenship: German • Visa Status: Green Card Holder

Skills

Technology Stack

- **Languages:** Python, PyData stack, C/C++, FORTRAN, ruby, lua, perl, HTML/CSS
- **Dev tools:** Git, Github, Docker, Portainer, make, cmake, conda, uv, Claude Code
- **HPC:** MPI, OpenMP, numba, BLAS/LAPACK, SLURM, Torque, Altair Grid Engine, Altair NavOps
- **Administration:** Ansible, Progress Chef, spack, Windows, MacOS, Linux (Ubuntu, Arch, RHEL)
- **Scientific software:** Schrodinger, Q-Chem, Gaussian, Orca, Cadpac, Mathematica, vmd, LaTeX

Computational and Technical

- 13 years of software development experience.
- 13 years of experience in data analysis and visualization.
- 13 years of experience in software development for HPC environments.
- 13 years of experience in version control and good software development practices.
- 8 years of experience with workstation and server-grade HPC hardware.
- 8 years of experience in system administration for HPC Linux systems.

Communication and Languages

- 12 years of experience presenting complicated scientific concepts to both specialist and general audiences.
- 12 years of scientific and technical writing experience for the purpose of publication and presentation.
- 8 years of experience in participating in and leading collaborations with teams from a variety of fields.
- Fluent in speaking, reading, and writing English and German. Intermediate conversational skills in Japanese.

Research and Teaching

- 13 years of experience combining analytical and theoretical work with computational methods.
- 13 years of experience teaching scientific and technical skills to specialist and general audiences.
- 12 years of independent development and pursuit of novel research avenues.
- 8 years of experience designing and leading research projects involving multiple team members.

Experience

Bristol Myers Squibb, Brisbane, CA, USA

Cheminformatics Systems Engineer

2023–present

Supporting and enabling computational drug design and machine learning within the Computational Sciences group. Ground up development and deployment of HPC infrastructure (Altair Grid Engine with NavOps cloud integration). Systems administration of diverse HPC environment, including integration with scientific software (Schrodinger, Python) and automated configuration (Ansible, spack, InfraChef). Enabling drug and scientific software development with cutting-edge AI tools (Claude Code, Anthropic, OpenAI and Gemini).

Kent State University, Kent, OH, USA

Postdoctoral Research Fellow working in the group of Prof. Barry Dunietz.

2022–2023

Scientific software development and application with a focus on the Q-Chem package for electronic structure. Applying screened range separated hybrid functionals within density functional theory to energy and charge transfer in biologically interesting systems.

University of Michigan, Ann Arbor, MI, USA

Postdoctoral Research Fellow working in the group of Prof. Eitan Geva.

2020–2023

Scientific software development and application of the improved dynamics methods to light-matter interactions. Work has been published in *The Journal of Physical Chemistry Letters* and *The Journal of Physical Chemistry C*. Purchase, installation and management of a cluster of high performance scientific workstations.

Swiss Federal Institute of Technology (ETH Zurich), Zurich, Switzerland

ETH Postdoctoral Fellow continuing the project with Prof. Jeremy Richardson.

2018–2020

Competitive 24-month postdoctoral fellowship financed through the European Commission (valued at \$250,000). Awarded based on a grant application including a detailed research proposal and interview panel.

Postdoctoral Researcher working in the group of Prof. Jeremy Richardson.

2017–2018

Development of improved quantum dynamics methods with a particular focus on computational efficiency. Leading a team of researchers in the development of a general-purpose, extendable scientific software package focused on scalability on high performance hardware. Work has been published in *The Journal of Chemical Physics* and *Faraday Discussions*.

Education

University of Warwick, Coventry, United Kingdom

PhD in Theoretical Chemistry

2013–2017

PhD Thesis: *Sampling minimal, adaptive basis sets for multidimensional, nuclear quantum dynamics using simple, semi-classical trajectories.*

Working in the group of Prof. Scott Habershon. Research was focused on development of new quantum dynamics methods with a particular focus on computational efficiency. After a rapid prototyping stage of theoretical work, focus shifted to scientific software development as well as optimization for HPC environments. Work has been published in *The Journal of Chemical Theory and Computation* and won the 2016 RSC Coulson Prize. Development Experience: Python, FORTRAN

Durham University, Durham, United Kingdom

MChem (Integrated Master's with Hons equivalent to a BS and MS) in Chemistry, 2:1

2009–2013

Masters thesis: *Electron affinities and overlap dispersion in density functional theory.*

Supervised by Prof. David Tozer. Investigated links between spurious overlap dispersion and trends in electron affinities and ionization potentials within density functional theory.

Publications

- Anisotropic Dielectric Screened Range-Separated Hybrid Density Functional Theory Calculations of Charge Transfer States across an Anthracene–TCNQ Donor–Acceptor Interface, C. Chakravarty, M.A.C. Saller, H. Aksu and B.D. Dunietz, *J. Chem. Theo. Comput.*, **20** 10751–10758 (2024)
- Cavity-modified Fermi's golden rule rate constants: Beyond the single mode approximation, M.A.C. Saller, Y. Lai and E. Geva, *J. Chem. Phys.*, **159** 151105 (2023)
- Cavity-Modified Fermi's Golden Rule Rate Constants from Cavity-Free Inputs, M.A.C. Saller, Y. Lai and E. Geva, *J. Phys. Chem. C*, **127** 3154–3164 (2023)
- Quasiclassical approaches to the generalized quantum master equation, G. Amati, M.A.C. Saller, A. Kelly and J.O. Richardson, *J. Chem. Phys.*, **23**, 234103 (2022)
- An Accurate Linearized Semiclassical Approach for Calculating Cavity-Modified Charge Transfer Rate Constants, M.A.C. Saller, Y. Lai and E. Geva, *J. Chem. Phys. Lett.*, **13**, 2330–2337 (2022)
- Benchmarking Quasiclassical Mapping Hamiltonian Methods for Simulating Cavity-Modified Molecular Dynamics, M.A.C. Saller, A. Kelly and E. Geva, *J. Chem. Phys. Lett.*, **12**, 3163–3170 (2021)
- Benchmarking Quasiclassical Mapping Hamiltonian Methods for Simulating Electronically Nonadiabatic Molecular Dynamics, X. Gao, M.A.C. Saller, Y. Liu, A. Kelly, J.O. Richardson and E. Geva, *J. Chem. Theo. Comput.*, **16**, 2883–2895 (2020)
- Path-integral approaches to nonadiabatic dynamics, Book Chapter in *Quantum Chemistry and Excited States: Methods and Applications*, M.A.C. Saller, J.E. Runeson and J.O. Richardson, Wiley, 629–653 (2020)
- Emerging opportunities and future directions: general discussion, *Faraday Discuss.*, **221**, 564–581 (2019)
- Quantum coherence in complex environments: general discussion, *Faraday Discuss.*, **221**, 168–201 (2019)
- Improved population operators for multi-state nonadiabatic dynamics with the mixed quantum-classical mapping approach, M.A.C. Saller, A. Kelly and J.O. Richardson, *Faraday Discuss.*, **221**, 150–167 (2019)
- On the identity of the identity operator in nonadiabatic linearized semiclassical dynamics, M.A.C. Saller, A. Kelly and J.O. Richardson, *J. Chem. Phys.*, **150**, 071101 (2019)
- Quantum Dynamics with Short-Time Trajectories and Minimal Adaptive Basis Sets, M.A.C. Saller and S. Habershon, *J. Chem. Theo. Comput.*, **13**, 3085–3096 (2017)
- Basis Set Generation for Quantum Dynamics Simulations Using Simple Trajectory-Based Methods, M.A.C. Saller and S. Habershon, *J. Chem. Theo. Comput.*, **11**, 8–16 (2015)

Awards and Grants

- **ETH Postdoctoral Fellowship 2018–2020** Competitive 24-month postdoctoral fellowship at ETH Zurich.
- **Warwick University Award for Teaching Excellence 2017** Postgraduate teaching commendation.
- **2016 Coulson Prize** Best graduate talk at the RSC Theoretical Chemistry Group Conference 2016.
- **2016 Contributed Talk Prize** RSC Spectroscopy and Dynamics Group Meeting 2016.
- **2016 Talk Prize** University of Warwick, Department of Chemistry Postgraduate Symposium 2016.

Open Source Projects

RSHTune.....

🔗 <https://github.com/maxsaller/RSHTune>

Range-separated hybrid density functionals are a user-tunable method in electronic structure theory, capable of screening long-range exchange in solution. The parameter controlling the separation between the long and short range is commonly optimised using DFT-Koopman's theorem. The RSHTune package automates the optimal tuning process for the popular Q-Chem electronic structure software package.

Conferences

- **52nd Midwest Theoretical Chemistry Conference**
Ohio State University, Jun 2022. Contributed talk
- **Faraday Discussion: Quantum effects in complex systems**
University of Warwick, Sep 2019. Presenting invited article and panel member for general discussion.
- **Recent developments in quantum dynamics, an E-CAM state-of-the-art workshop**
Lyon, June 2019
- **C4 Computational Workshop**
University of Zurich, June 2019. Invited talk
- **Computational Molecular Science**
University of Warwick, March 2019
- **CECAM Workshop: Non-adiabatic quantum dynamics: From theory to experiment**
University of Lausanne, July 2018

Teaching and Service

- **Lecture Courses, University of Michigan**
Teaching undergraduate lecture courses, CHEM 262 - Mathematical Methods for Physical Scientists and CHEM 461 - Physical Chemistry I (Quantum Mechanics), on a substitute basis.
- **Mathematica Workshops, ETH Zurich**
Developing and teaching an introductory undergraduate workshop for the Mathematica software and Wolfram language. Focus on numerical and analytical approaches to scientific problems.
- **Programming Skills for Chemists, ETH Zurich**
Introduction to programming in C/C++, compilation and working in Linux. Focus on establishing programming as a tool first year students can use to solve problems over the course of their degree.
- **Physical Chemistry Tutorials, ETH Zurich**
Leading problem classes with small groups of undergraduate students. Working through topics covered in lectures, contextualizing and deepening understanding. Providing support solving typical exam problems.
- **Masters Research Projects, University of Warwick**
Coordinating and supervising 6-month undergraduate research projects. Introduction to programming, working in high-performance computing environments and scientific writing ($\text{\LaTeX}$).
- **Computational Chemistry Laboratory Projects, University of Warwick**
Supervising and developing undergraduate computer laboratory projects. Crash course in Linux and terminal environments. Introduction to computational chemistry software culminating in a research project.
- **Spectroscopy in Schools, University of Warwick/Royal Society of Chemistry**
Providing high school students with hands-on experience using portable, research-grade spectroscopy equipment. Providing advice on higher education, particularly in STEM fields.